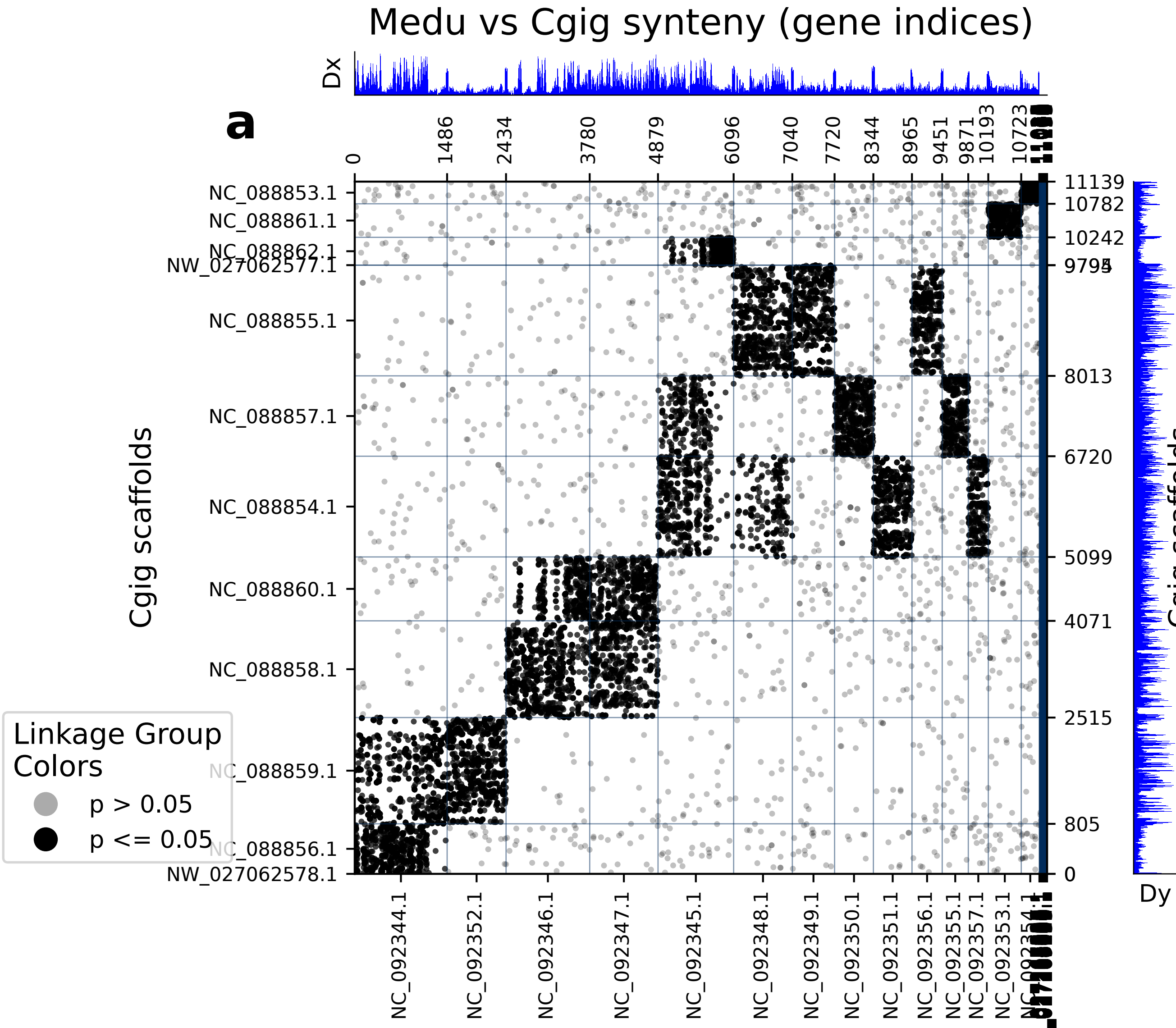
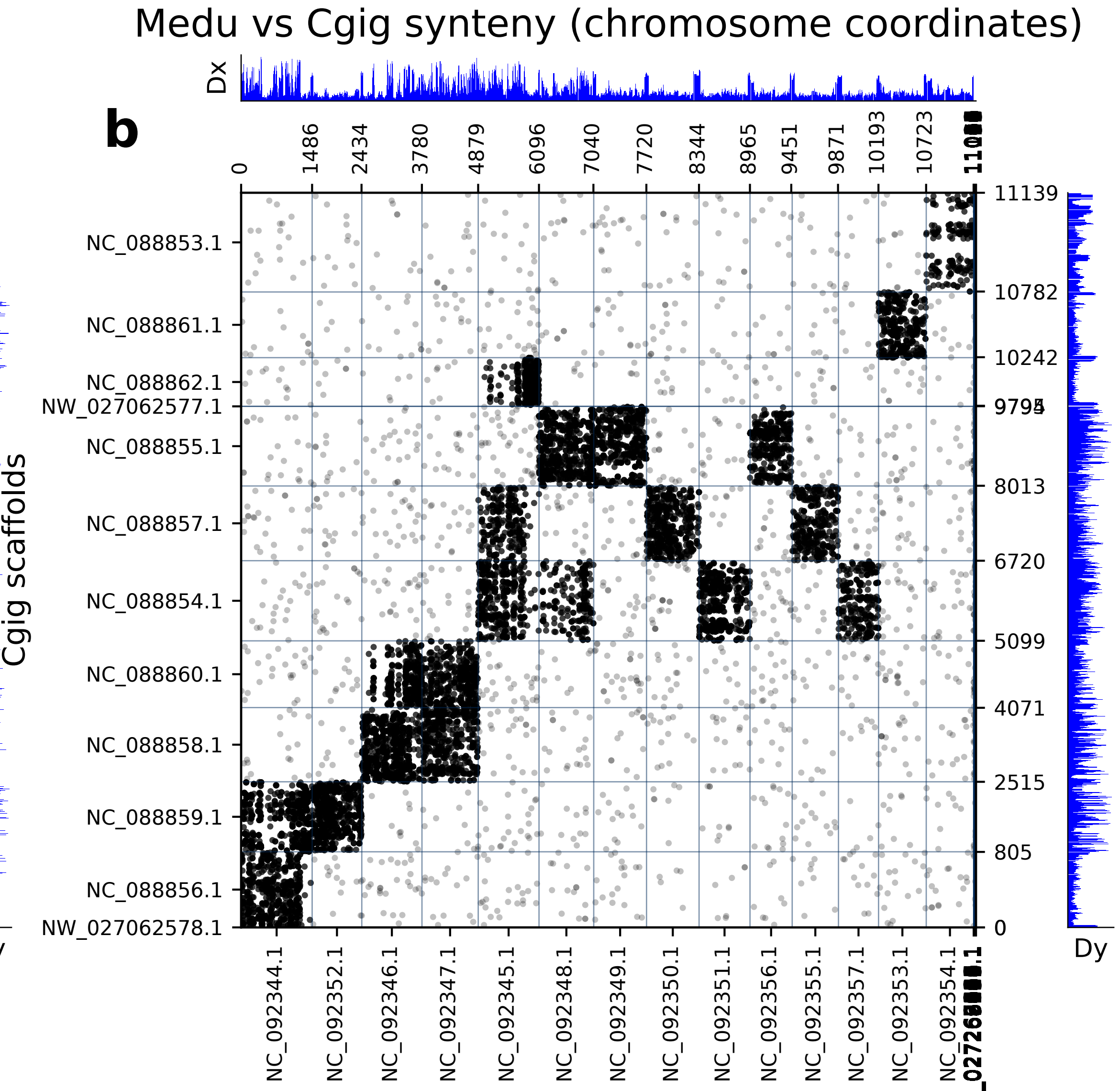
diamond: 11140 orthologs on 129 Medu scaffolds and 12 Cgig scaffolds.





c

Medu vs Cgig groups

Medu scaffolds

d

Medu vs Cgig FET, whole chroms.

	Medu scaf	Cgig scaf	p (FET)	Count
0	NC_092352.1	NC_088859.1	0.00e+00	840
1	NC_092346.1	NC_088858.1	0.00e+00	840
2	NC_092344.1	NC_088859.1	4.92e-248	729
3	NC_092348.1	NC_088855.1	1.91e-309	635
4	NC_092344.1	NC_088856.1	0.00e+00	612
5	NC_092349.1	NC_088855.1	0.00e+00	545
6	NC_092347.1	NC_088858.1	5.30e-192	536
7	NC_092351.1	NC_088854.1	0.00e+00	525
8	NC_092350.1	NC_088857.1	0.00e+00	509
9	NC_092347.1	NC_088860.1	7.42e-206	451
10	NC_092345.1	NC_088854.1	1.86e-79	427
11	NC_092353.1	NC_088861.1	0.00e+00	393
12	NC_092356.1	NC_088855.1	6.10e-212	379
13	NC_092346.1	NC_088860.1	2.63e-100	375
14	NC_092345.1	NC_088862.1	1.71e-262	352
15	NC_092355.1	NC_088857.1	2.07e-237	332
16	NC_092345.1	NC_088857.1	1.89e-21	258
17	NC_092357.1	NC_088854.1	4.14e-140	246
18	NC_092348.1	NC_088854.1	5.86e-08	209
19	NC_092354.1	NC_088853.1	1.80e-214	183

e

Medu vs Cgig FET, chrom. pieces

	Medu scaf	Cgig scaf	p (FET)	Count
0	NC_092352.1 : 0-end	NC_088859.1 : 0-end	0.00e+00	840
1	NC_092346.1 : 0-end	NC_088858.1 : 0-end	0.00e+00	840
2	NC_092344.1 : 0-end	NC_088859.1 : 0-end	4.92e-248	729
3	NC_092348.1 : 0-end	NC_088855.1 : 0-end	1.91e-309	635
4	NC_092344.1 : 0-end	NC_088856.1 : 0-end	0.00e+00	612
5	NC_092349.1 : 0-end	NC_088855.1 : 0-end	0.00e+00	545
6	NC_092347.1 : 0-end	NC_088858.1 : 0-end	5.30e-192	536
7	NC_092351.1 : 0-end	NC_088854.1 : 0-end	0.00e+00	525
8	NC_092350.1 : 0-end	NC_088857.1 : 0-end	0.00e+00	509
9	NC_092347.1 : 0-end	NC_088860.1 : 0-end	7.42e-206	451
10	NC_092345.1 : 0-end	NC_088854.1 : 0-end	1.86e-79	427
11	NC_092353.1 : 0-end	NC_088861.1 : 0-end	0.00e+00	393
12	NC_092356.1 : 0-end	NC_088855.1 : 0-end	6.10e-212	379
13	NC_092346.1 : 0-end	NC_088860.1 : 0-end	2.63e-100	375
14	NC_092345.1 : 0-end	NC_088862.1 : 0-end	1.71e-262	352
15	NC_092355.1 : 0-end	NC_088857.1 : 0-end	2.07e-237	332
16	NC_092345.1 : 0-end	NC_088857.1 : 0-end	1.89e-21	258
17	NC_092357.1 : 0-end	NC_088854.1 : 0-end	4.14e-140	246
18	NC_092348.1 : 0-end	NC_088854.1 : 0-end	5.86e-08	209
19	NC_092354.1 : 0-end	NC_088853.1 : 0-end	1.80e-214	183

(a) depicts the chromosomes/scaffolds of Medu (x-axis) plotted against the chromosomes/scaffolds of Cgig (y_axis). Each dot in the plot represents an ortholog, specifically a reciprocal best diamond blastp match between two species. The unit of the x- and y-axes are the number of orthologous proteins between these two species: 11140 orthologs found between 129 Medu scaffolds and 12 Cgig scaffolds. If there are chromosome breaks, Fisher's exact test is used to calculate the significance of the interactions between the sub-chromosomal pieces. Otherwise Fisher's exact test is calculated on whole chromosomes. The opacity of the dots depict the significance from Fisher's exact test. Dots that are a solid color are in cells with a FET p-value less than or equal to 0.05. Dots that are translucent are in cells with a FET p-value greater than 0.05. The Dx and Dy values depict places where there may be sudden breaks in synteny. See the supplementary information the following paper for more information on Dx and Dy: Simakov, Oleg, et al. Deeply conserved synteny resolves early events in vertebrate evolution. Nature Ecology & Evolution 4.6 (2020): 820-830. (b) depicts the same information as panel a, but it is plotted in the organisms chromosome basepair coordinates rather than gene index. This is useful for visualizing gene-poor regions of the chromosomes (c) shows which gene groups are most prevalent in each chromosome pair. The FET p-value in this table corresponds to the whole-chormosome FET p-value , and is not a FET value of the correlation between the chromosome pair and the gene group. (d) shows chromosome-scale significance values. This table shows the same information as c, but does not factor in gene group information. (e) shows the FET p-values of the sub-chromosomal compartments. Useful for showing if single arms of chromosomes are correlated with other regions.